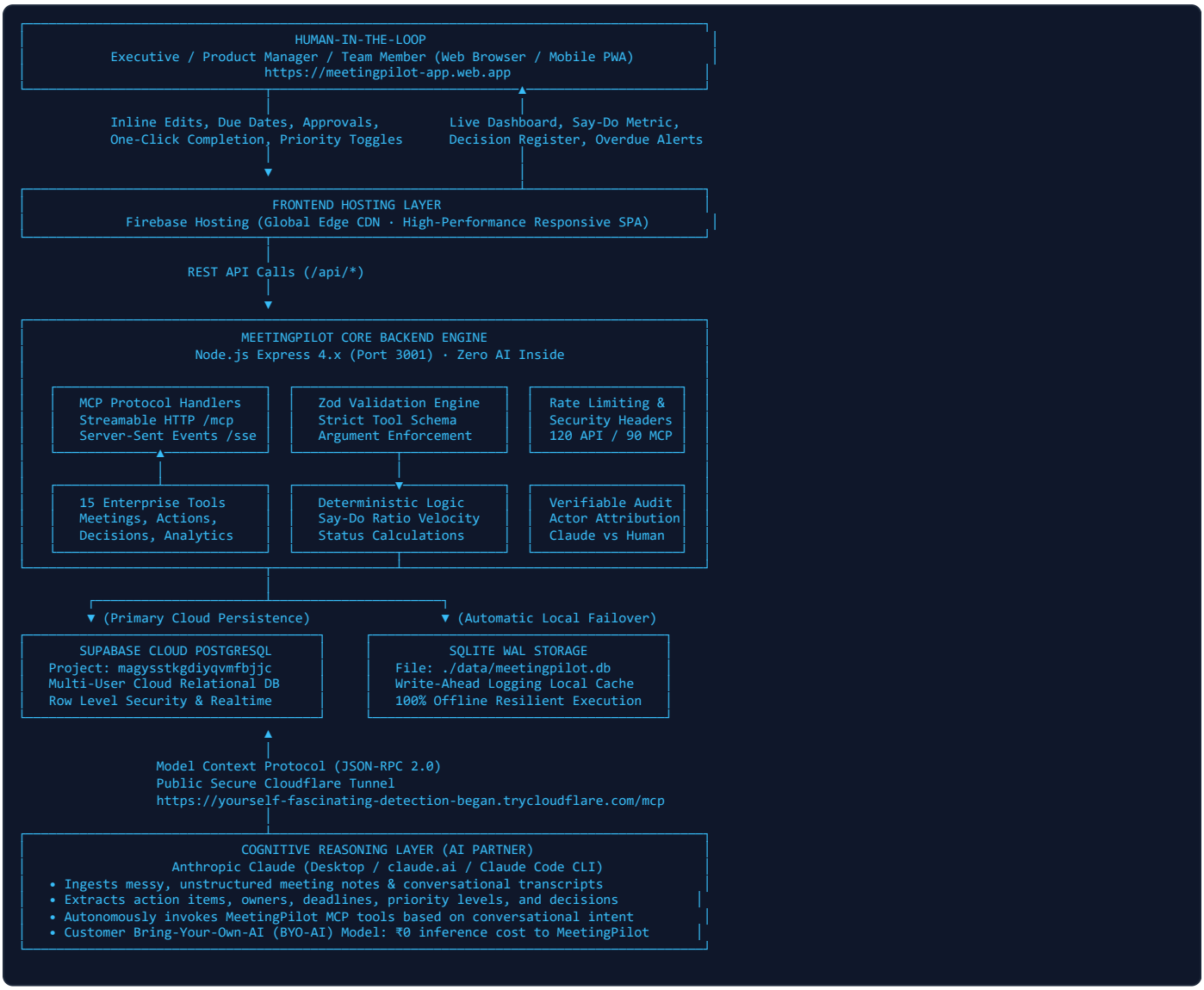
1. HIGH-LEVEL ARCHITECTURE TOPOLOGY

Clean architectural decoupling: Claude provides the cognitive reasoning layer via Model Context Protocol (MCP), while MeetingPilot executes 100% deterministic persistence with zero AI code inside.



2. END-TO-END DATA FLOW (10-STEP OPERATIONAL LIFECYCLE)

Step 1 to 5: Conversational Ingestion & Execution

- **1. Conversational Input:** User pastes meeting transcript or bulleted notes into Claude.
- **2. Contextual Parsing:** Claude reasons over text, disambiguating tasks, owners, and decisions.
- **3. MCP Tool Invocation:** Claude formulates JSON-RPC 2.0 tool calls (`create_meeting`, `add_action_item`).
- **4. Secure Edge Transit:** Transmitted via Streamable HTTP `POST /mcp` over Cloudflare edge tunnel.
- **5. Schema Validation:** MeetingPilot validates arguments against strict Zod type definitions.

Step 6 to 10: Persistence, Oversight & Closed-Loop

- **6. Atomic Persistence:** Record stored in Supabase PostgreSQL (or local SQLite WAL fallback).
- **7. Audit Attribution:** State change recorded in `audit_logs` attributing `"claude_mcp"`.
- **8. Live UI Reflection:** Dashboard (`meetingpilot-app.web.app`) renders items in real time.
- **9. Human Veto & Edit:** Human adjusts due date, reassigns owner, or clicks task completed.
- **10. Closed-Loop AI Reflection:** When asked, Claude queries `get_dashboard_summary` and reflects the user's edits.

3. DIVISION OF RESPONSIBILITIES: VOLOPAY VS. PARTNERS

Entity / Layer	Primary Component	Scope of Responsibilities
Volopay / MeetingPilot Core	System of Record	Manages meetings, actions, Decision Register, Say-Do ratio velocity math, audit trail, and responsive web dashboard.
Volopay Platform Integration	Fintech Governance	Captures meeting-related financial commitments (`create_spend_approval`) and routes to corporate card/budget rails.
AI Partner (Anthropic Claude)	Cognitive Intelligence	Speech/text comprehension, intent extraction, priority assignment, and autonomous Model Context Protocol orchestration.
Database Partner (Supabase)	Cloud Persistence	Managed PostgreSQL cloud instance, connection pooling, automated backups, and row-level security.
Edge Partner (Firebase)	Frontend Delivery	Global Edge CDN hosting, SSL termination, and static asset distribution (`meetingpilot-app.web.app`).
Network Partner (Cloudflare)	Secure Edge Ingress	Zero-trust QUIC/HTTPS tunnel exposing the local microservice without open inbound firewall ports.

4. TOOLS & TECHNOLOGIES USED

Runtime & MCP Protocol

- **Node.js v24.14 LTS:** Asynchronous engine.
- **Express 4.x:** REST API & security middleware.
- **@modelcontextprotocol/sdk:** Official MCP implementation (Streamable HTTP & SSE).
- **Zod v3.25:** Strict runtime argument validation.

Databases & Persistence

- **Supabase Cloud PostgreSQL:** Multi-user cloud relational persistence.
- **better-sqlite3:** Write-Ahead Logging (WAL) local database for zero-downtime offline fallback.
- **Audit Trail Engine:** Verifiable attribution.

Frontend & Tooling

- **Firebase Edge Hosting:** SPA hosting.
- **Vanilla JavaScript & CSS3:** Zero framework latency, light/dark themes.
- **pptxgenjs & docx:** Programmatic pitch deck and Word document synthesis.

5. ARCHITECTURAL ASSUMPTIONS

- **Bring Your Own AI (BYO-AI):** Customers provide their own Claude accounts (Desktop/Web/CLI). MeetingPilot pays ₹0 in inference fees, maintaining 97.5%+ gross margins.
- **Deterministic Relational Storage:** Business records (deadlines, owners, priority, status) must live in relational SQL, eliminating LLM hallucination in corporate data.
- **Human Agency Priority:** AI proposes and queries commitments; human users retain unconditional authority to edit, postpone, reassign, or cancel tasks.
- **Multi-Tenant Workspace Partitioning:** Records are logically grouped by Workspace (Product & Engineering, Leadership, Sales) for organizational isolation.

6. DISTINGUISHING WHAT WAS BUILT VS. PROPOSED PRODUCTION SYSTEM

Dimension	What We Built (Live Working System)	Proposed Production System (Scale-Out)
MCP Toolset	**15 Fully Functional Tools** covering meetings, actions, decisions, and analytics; verified in Claude.	15 Tools + Automated Background Worker for scheduled deadline escalation & reminders.
Authentication	**Bearer Token Auth** (`MCP_API_TOKEN`) with public evaluation demo mode.	**OAuth 2.0 / OpenID Connect** with user-delegated tokens and SAML SSO (Okta / Azure AD).
Database	**Dual-Engine:** Supabase PostgreSQL cloud instance + automatic SQLite WAL offline fallback.	Multi-tenant PostgreSQL with isolated customer schemas and automated geo-replication.
Deployment	**Firebase Edge Hosting** (Frontend) + **Cloudflare Tunnel** (Backend microservice).	**Google Cloud Run / Kubernetes (EKS)** auto-scaling containers behind Cloudflare Load Balancers.
Rate Limiting	In-memory sliding-window rate limiter (120 API / 90 MCP req/min).	**Distributed Redis Cluster** (Upstash / ElastiCache) across availability zones.
Volopay Integration	Spend approval tracking tools (`create_spend_approval`, `approve_action`).	**Direct Volopay Core API Bridge:** Automated virtual card provisioning upon meeting approval.
Integrations	Outgoing webhook dispatch endpoints.	**Native Bi-Directional Slack & MS Teams Apps** with interactive approval buttons.

7. Prototype Limitations

- **Shared Token Scope:** Uses a shared bearer token rather than per-seat OAuth 2.0 PKCE user authorization.
- **In-Memory Rate Limiting:** Rate limit state is in-process; horizontal scaling requires centralized Redis.
- **Manual Audio Input:** Processes transcripts/notes pasted into Claude; continuous audio streaming requires external STT.

8. Next Steps & Development Roadmap

- **Phase 1 (Q1 2027): Volopay Card Bridge:** Issue single-use virtual cards for approved meeting budgets; add OAuth 2.0.
- **Phase 2 (Q2 2027): Omnichannel Sync:** Bi-directional Slack & Teams bots; Google/Outlook calendar sync.
- **Phase 3 (Q3 2027): Agenda Rollover:** Automatically roll incomplete tasks into next week's meeting agenda.

Live Production Links & Verification Artifacts

Live Web App: meetingpilot-app.web.app

Live MCP Endpoint:
`...trycloudflare.com/mcp`

GitHub Codebase:
applicationmanifester/MeetingPilot